2. Threat Intelligence Integration Documentation

The objective of this task was to integrate external threat intelligence with the SIEM platform to enrich security alerts and perform threat hunting activities. Threat intelligence feeds help analysts identify malicious indicators such as suspicious IP addresses, domains, or file hashes.

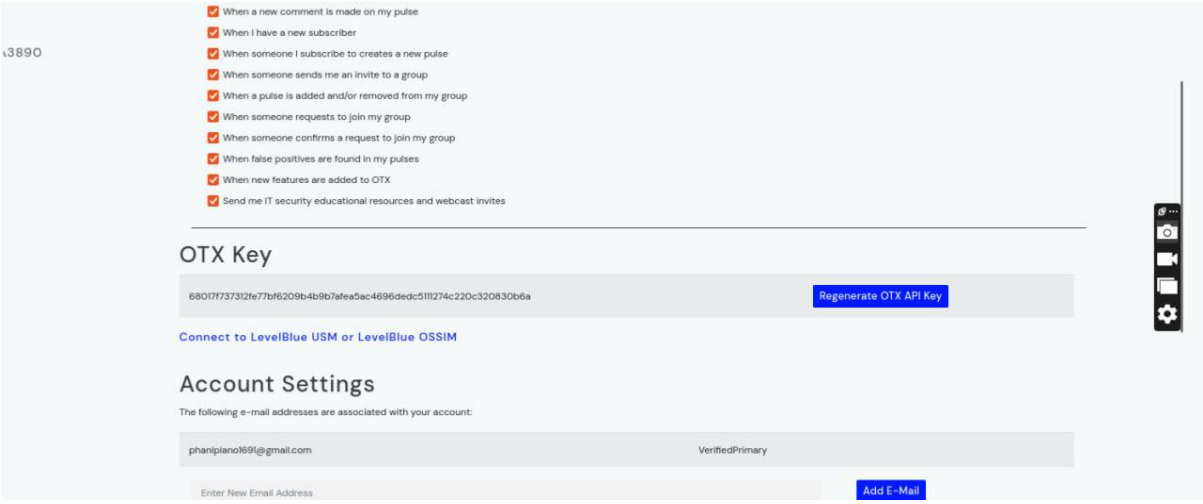
1. Tools Used

- Wazuh
- AlienVault OTX
- TheHive

2. Threat Feed Import

An account was created on AlienVault OTX and an API key was generated.

The API key was added to the Wazuh configuration file.



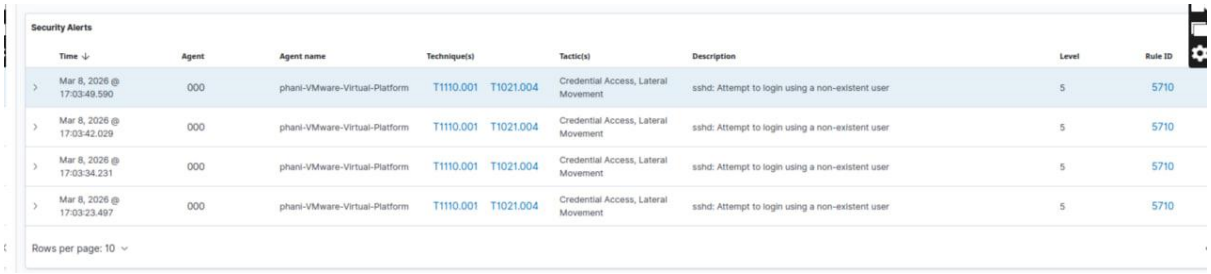
Configuration file:

nano /var/ossec/etc/ossec.conf

Integration configuration:

```
<integration>
  <name>alienvault</name>
```

```
  <api_key>OTX_API_KEY</api_key>
  <alert_format>json</alert_format>
</integration>
```



Security Alerts

Time	Agent	Agent name	Technique(s)		Tactic(s)	Description	Level	Rule ID
Mar 8, 2026 @ 17:03:49.590	000	phani-VMware-Virtual-Platform	T1110.001	T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Mar 8, 2026 @ 17:03:42.029	000	phani-VMware-Virtual-Platform	T1110.001	T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Mar 8, 2026 @ 17:03:34.231	000	phani-VMware-Virtual-Platform	T1110.001	T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Mar 8, 2026 @ 17:03:23.497	000	phani-VMware-Virtual-Platform	T1110.001	T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710

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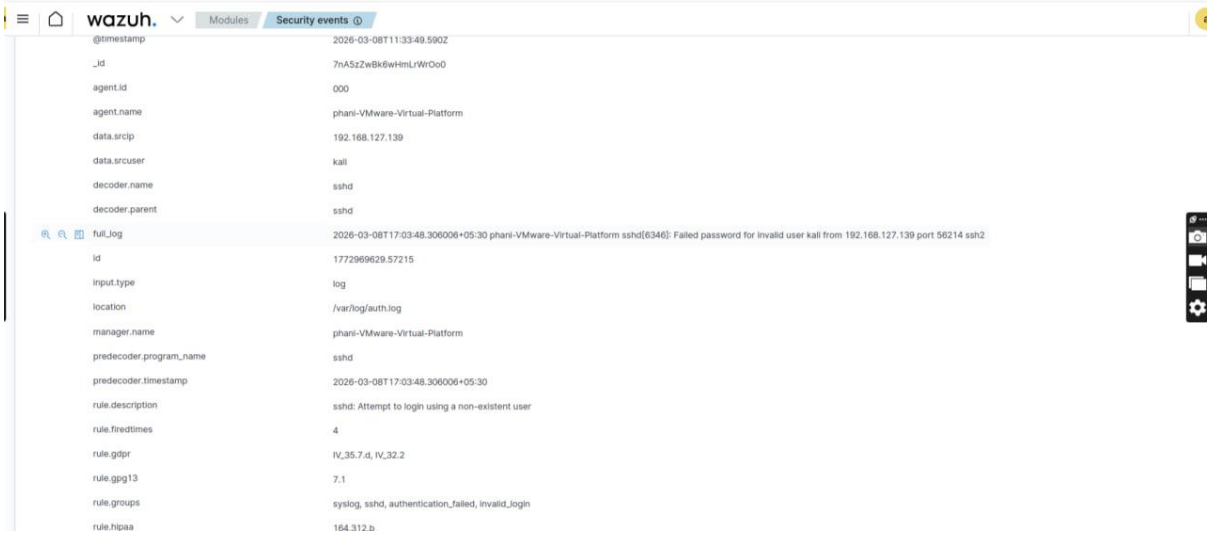
After adding the integration, the Wazuh service was restarted to apply the configuration.
Command used:
sudo systemctl restart wazuh-manager

This allowed Wazuh to query AlienVault OTX feeds when processing alerts.

3. IOC Testing

A mock IP address was used to simulate an indicator of compromise.
Test IP:

192.168.127.139



wazuh. Modules Security events

@timestamp	2026-03-08T11:33:49.590Z
_id	7nA5zZwBk6wHmLrWrOo0
agent.id	000
agent.name	phani-VMware-Virtual-Platform
data.srcip	192.168.127.139
data.srcuser	kali
decoder.name	sshd
decoder.parent	sshd
full_log	2026-03-08T17:03:48.306006+05:30 phani-VMware-Virtual-Platform sshd[6346]: Failed password for invalid user kali from 192.168.127.139 port 56214 ssh2
id	1772969629.57215
input.type	log
location	/var/log/auth.log
manager.name	phani-VMware-Virtual-Platform
predecoder.program_name	sshd
predecoder.timestamp	2026-03-08T17:03:48.306006+05:30
rule.description	sshd: Attempt to login using a non-existent user
rule.firedtimes	4
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	7.1
rule.groups	syslog, sshd, authentication_failed, invalid_login
rule.hipaa	164.312.b

This IP was searched in the Wazuh security events dashboard to check whether it matched any threat intelligence indicators from OTX.

No malicious matches were detected in the available logs.

6. Alert Enrichment

Threat intelligence enrichment allows alerts to be supplemented with reputation data from external intelligence feeds.

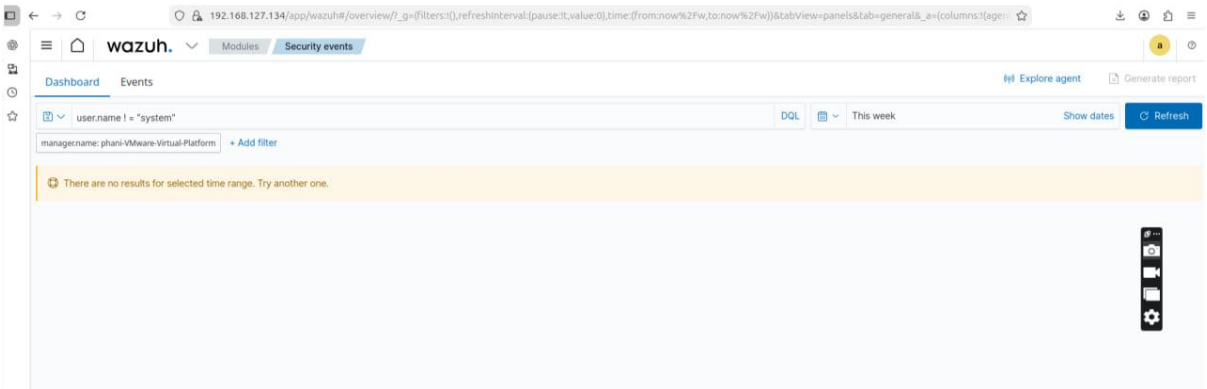
Alert ID	IP	Reputation	Notes
003	192.,168.127.134	Not Found	No threat intelligence match

The result indicates that the tested IP address was not identified as malicious in the OTX database.

5. Threat Hunting

Threat hunting was performed to detect suspicious login activity associated with MITRE ATT&CK Technique T1078 – Valid Accounts.
Query used in Wazuh:

NOT user.name:"system"



This query filters authentication logs and excludes system accounts to identify activity performed by normal user accounts.
During the investigation, no abnormal login behavior was detected in the collected logs.

Threat intelligence integration was implemented by connecting Wazuh with AlienVault OTX threat feeds. Indicators of compromise were evaluated against the threat intelligence database. A test IP address was analyzed but no malicious reputation was found. Threat hunting was performed on authentication logs to detect potential misuse of valid user accounts